MIMO Beamforming with Reduced Pilots and Radio Map for Emergency Wireless Communications

Bin Yang*, Hong Hao Zhang†
*University of New South Wales, Sydney, NSW 2052, Australia
†Shenzhen College of International Education, Shenzhen 518048, China
Email: bin.yang@unsw.edu.au

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***Abstract*—In large social gatherings or emergencies, the increasing communication demands can lead to congestion of the network, resulting in some user equipment (UE) being out of service. In such situations, it is crucial to provide emergency communication service to ensure reliable connectivity for all UEs. A key to this service is the rapid and accurate acquisition of channel state information (CSI) for beamforming, which is typically obtained by pilot-based methods. However, pilot-based methods may degrade seriously due to overhead and contamination. To address this issue, in this paper we consider reducing the pilot overhead and incorporating radio map to assist pilot-based beamforming. The prior location information provided by the radio map can reduce pilot overhead and help to counteract the loss caused by pilot contamination. According to the numerical simulations, beamforming with reduced pilots is superior to that with full pilots and can be further improved by integrating with the radio map.**

***Index Terms*—Beamforming, deep learning, emergency communications, radio map.**

I. Introduction

In large social gatherings or emergencies, the communication demands grow rapidly compared to normal periods. For example, after the 512 Wenchuan earthquake in China, the communication needs of users in Sichuan surged to ten times their ordinary levels, while the demands from the rest of China to Sichuan increased by 5 to 6 times [1]. The massive increase in communication demands poses a significant challenge to existing telecommunication networks [2], resulting in some user equipment (UE) being out of service [3]. In such situations, it is crucial to provide emergency communication service [4] to ensure reliable connectivity for as many UEs as possible [5]. The emergency communications usually require accurate channel state information (CSI) for beamforming to ensure the high-reliable and high-speed communication link between the base station (BS) and the UE [6]. In current wireless communications, pilots are utilized to acquire the CSI. However, pilot-based beamforming has many issues, especially in emergency communications. To be specific, pilot overhead increases with the number of antennas, which occupies large time-frequency resources for data transmission. This is unacceptable especially during emergencies. In addition, pilots are vulnerable to interference and noise, resulting in poor performance of channel estimation.

Various technologies are studied to provide high-reliable and high-speed communication links in emergencies. In [7], an energy minimization-based beam training optimization problem in reconfigurable intelligent surfaces (RIS)-assisted unmanned aerial vehicle (UAV) emergency communication system is considered. In [8], a resource scheduling scheme has been developed to maximize the sum spectral efficiency (SE) for half-duplex UAV-assisted emergency communication networks. In [9], the channel capacity is enhanced in emergency scenarios through cognitive radio and spectrum sharing research. Radio map [10]–[13] is also a promising technology to enhance the reliability and improve the rate of emergency communications. Initially, radio map is defined as the geographical signal power spectrum density as a function of location, frequency, and time [10]. Then, it has been expanded to more channel-related information like pathloss, time of arrival (ToA), phase, azimuth, and zenith angle of departure (AoD), and angle of arrival (AoA) [11]. In [12], [13], radio map has been further customized to be task-oriented, capable of directly generating beamforming vector based on UE's location information.

This paper introduces location information to reduce the pilot overhead and help counteract the loss caused by pilot contamination. The contributions of this paper can be summarized as follows.

- We propose an end-to-end method for beamforming with reduced pilots based on deep learning. The designed neural network can perform joint extrapolation, denoising, and beamforming with the reduced pilots. The trade-off between overhead and performance is investigated.
- We study the performance characteristics of the beamforming schemes based on radio map and reduced pilots. To leverage the strengths and compensate for the weaknesses of these two approaches, we integrate the beamforming vectors obtained from the reduced pilots with those generated by radio map.
- We validate the superiority of the reduced pilots-based beamforming empowered by deep learning compared to the conventional full pilot scheme. The integration of reduced pilots and radio map can effectively elevate both the performance floor and ceiling according to the simulation results.

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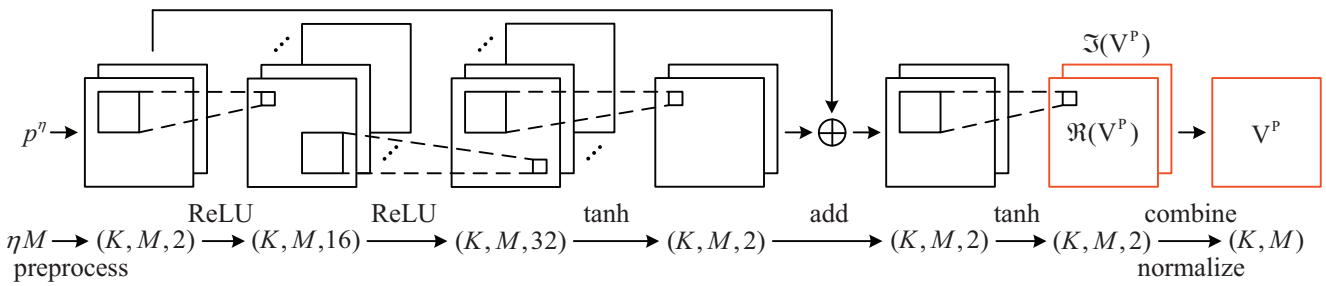

Fig. 1: The framework of reduced pilots-based beamforming.

The remainder of the paper is organized as follows. In Section II, the system model for emergency communications is presented. In Section III, the deep learning-based reduced pilots method for beamforming is proposed. In Section IV, an integration strategy of radio map and reduced pilots has been proposed. In Section V, numerical simulations and results are given, followed by conclusions in Section VI.

Notations: a, $\mathbf{a}$ and $\mathbf{A}$ represent a scalar, a vector, and a matrix, respectively. $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$ and $(\cdot)^\dagger$ represent conjugate, transpose, conjugate transpose and Moore-Penrose inverse operation, respectively. $\mathcal{CN}(\mu,\sigma^2)$ represents complex Gaussian distribution with the mean μ and the standard deviation σ. $\mathbb{R}$ and $\mathbb{C}$ stand for the set of real and complex numbers, respectively. $\Re(\cdot)$ and $\Im(\cdot)$ denote the real and imaginary parts of the complex number, respectively. $\odot$ and $\oslash$ denote the element-wise (Hadamard) product and division, respectively.

II. System Model

We consider a single-cell multi-input single-output (MISO)-orthogonal frequency division multiplexing (OFDM) downlink transmission scenario for the emergency communications where a BS equipped with M transmit antennas serves the single-antenna UE over K subcarriers. The number of samples for the UE at different locations is L, and the location of UE is available. The received signal of the UE over the kth subcarrier can be written as

$$y_k = \mathbf{h}_k^H \mathbf{x}_k + w_k = \mathbf{h}_k^H \mathbf{v}_k s_k + w_k, \tag{1}$$

where $\mathbf{h}_k \in \mathbb{C}^{M\times 1}$ is the downlink CSI, $\mathbf{x}_k = \mathbf{v}_k s_k \in \mathbb{C}^{M\times 1}$ is the transmitted signal where $\mathbf{v}_k \in \mathbb{C}^{M\times 1}$ is the beamforming vector and s_k is a complex number representing the transmitted symbol, $w_k \sim \mathcal{CN}(0,\sigma^2)$ is the complex additive Gaussian noise. The received signal-to-noise ratios (SNR) has the form

$$r_k = \frac{\left|\mathbf{h}_k^H \mathbf{v}_k\right|^2}{\sigma^2}. \tag{2}$$

The SE for the data transmission of the UE over all subcarriers can be averaged to

$$R = \frac{1}{K}\sum_{k=1}^{K} \log\left(1+r_k\right)\left(1-\rho\right), \tag{3}$$

where ρ represents the proportion of the pilot overhead and can be calculated as the ratio of time-frequency resources used for pilot transmission to total resources.

III. Reduced Pilots-Based Beamforming

The designed reduced pilots-based beamforming generates the beamforming vector $\mathbf{V}_l^\mathrm{P}$ of the UE at the lth location by taking its received pilots $\boldsymbol{p}_l^\eta$ as input, i.e.,

$$\mathbf{V}_l^\mathrm{P} = \mathcal{F}\left(\boldsymbol{p}_l^\eta;\theta_\mathrm{P}\right), \tag{4}$$

where $\boldsymbol{p}_l^\eta = [Y_{l,1};Y_{l,2};\cdots;Y_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ and the superscript η is the proportion of antennas transmitting pilots to the total number of antennas at the BS, $Y_{l,i}$ represents the received pilot symbol of the UE at the lth location, and θ_P is the set of the parameters of the neural network for reduced pilots-based beamforming.

The framework of the proposed reduced pilots-based beamforming is shown in Fig. 1. As shown in the figure, the received pilots need to be preprocessed to adapt to the input dimensions of the neural network. The first step involves channel estimation based on the received pilots. The channel gains of the antennas transmitting pilots to the UE can be obtained by least squares (LS) estimation as

$$\mathbf{h}_l^\eta = \mathbf{p}_l^\eta \oslash \mathbf{x}_l^\eta, \tag{5}$$

where $\mathbf{h}_l^\eta = [h_{l,1};h_{l,2};\cdots;h_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ is the estimated channel gains of the antennas that transmitting pilots, $\mathbf{x}_l^\eta = [X_{l,1};X_{l,2};\cdots;X_{l,\eta M}] \in \mathbb{C}^{\eta M\times 1}$ represents the transmitted pilot symbols. Then, $\mathbf{h}_l^\eta$ is normalized to meet the constraint of $||\mathbf{h}_l^\eta|| = 1$. After that, zero-padding is performed on the gains of the remaining antennas to increase the dimension from ηM to M. The channel gains after padding with zeros is denoted as $\mathbf{h}_l' \in \mathbb{C}^{M\times 1}$, which can be further extended to $\mathbf{H}_l' = \left[\mathbf{h}_l',\mathbf{h}_l',\cdots,\mathbf{h}_l'\right]^H \in \mathbb{C}^{K\times M}$ by repeating K times of $\mathbf{h}_l'$. The matrix $\mathbf{H}_l'$ can be seen as the partial CSI, and joint channel estimation and beamforming need to be performed for the beamforming vector. Finally, $\mathbf{H}_l'$ is decomposed into real and imaginary parts before being fed to the neural network.

Then, the task of the neural network is to generate the beamforming vector by jointly performing extrapolation, denoising, and beamforming with the preprocessed data. CNN is selected due to its success in image super-resolution. In Fig. 1, the preprocessed data with dimensions $(K,M,2)$ is fed into the neural network. The convolutional kernel size of the CNN is 3×3, which is a standard size in the field of image processing. The stride and zero-padding are both set to 1 to ensure that

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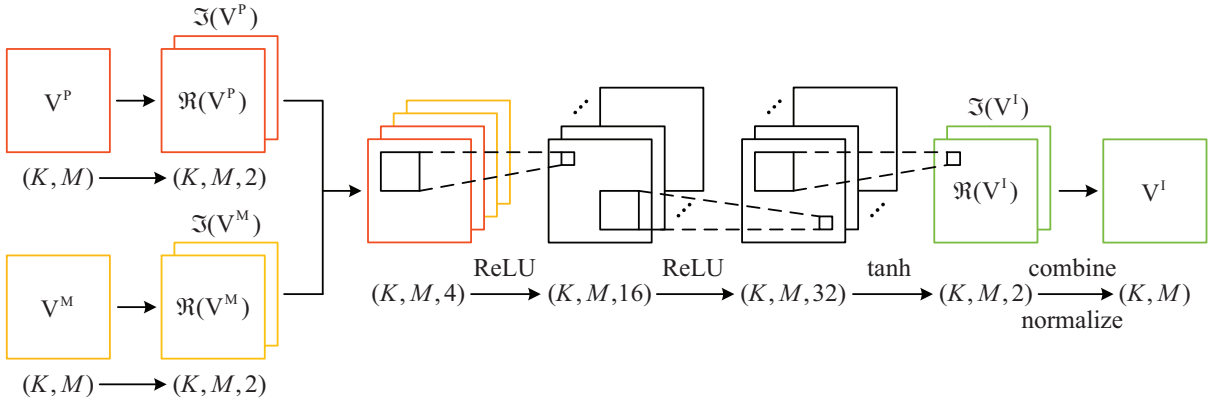

Fig. 2: The framework for the integration of reduced pilots and radio map.

the output size remains unchanged. Two convolutional layers are employed to gradually increase the depth from 2 to 32 to extract more features of the input data. Rectified linear unit (ReLU) is applied here to activate neurons, introducing non-linearity to the model without significantly affecting the computational complexity. After that, these high-level features are aggregated to match the dimensions expected by the final output data, and hyperbolic tangent (tanh) is applied as the activation function. To enhance the performance of the neural network, additional layers can be added on this basis, and residual network (ResNet) can be applied to prevent the degradation problem. As shown in Fig. 1, a residual connection is introduced to add the input to the output. After the addition operation, the last convolutional layer serves to map the output range to $(-1,1)$ by utilizing the tanh activation function. Then, two channels of the tensor are combined into complex data with the dimension of (K,M), followed by normalization to obtain the generated beamforming vector $\mathbf{V}^{\mathrm{P}}$.

To train the built neural network, the loss function is customized to be the negative of the downlink SE in the scenario of single UE without taking interference into account as in [12].

$$\begin{aligned}&\mathcal{L}\Big(\mathcal{F}\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right),\mathbf{H}_l\Big)\\&=-\mathcal{R}\Big(\mathcal{F}\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right),\mathbf{H}_l\Big)\\&=-\frac{1}{K}\sum_{k=1}^{K}\log\left(1+\frac{\left|\mathbf{h}_{l,k}^{H}\mathcal{F}_k\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right)\right|^2}{\sigma^2}\right),\end{aligned}\tag{6}$$

where $\mathbf{H}_l=[\mathbf{h}_{l,1},\mathbf{h}_{l,2},\cdots,\mathbf{h}_{l,K}]^H\in\mathbb{C}^{K\times M}$ is the downlink CSI for the UE at the lth location and $\mathbf{h}_{l,k}\in\mathbb{C}^{M\times 1}$ denotes the channel gain of the kth subcarrier, $\mathcal{G}_k\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right)\in\mathbb{C}^{M\times 1}$ is the generated beamforming vector over the kth subcarrier. The training objective is to minimize the expected loss function of the UE at all L locations, i.e.,

$$\theta_{\mathrm{P}}^{*}=\arg\min_{\theta_{\mathrm{P}}}\frac{1}{L}\sum_{l=1}^{L}\mathcal{L}\Big(\mathcal{F}\left(\boldsymbol{p}_l^{\eta};\theta_{\mathrm{P}}\right),\mathbf{H}_l\Big).\tag{7}$$

It has been proved in [12] that the customized loss function in (6) is differentiable with respect to θ_{P}, which provides feasibility for gradient calculation and weight update.

IV. The Integration of Radio Map and Reduced Pilots

Pilot and location are two different data sources, leading to different performances of the beamforming schemes based on them. To leverage the strengths and compensate for the weaknesses of these two methods, we integrate location with reduced pilots for better performance in varying environments.

A. Radio Map-Based Beamforming

Radio map establishes a connection between UE's location and its CSI, and radio map-based beamforming generates the beamforming vector $\mathbf{V}_l^{\mathrm{M}}$ for the UE by taking its location $\boldsymbol{g}_l$ as input, i.e.,

$$\mathbf{V}_l^{\mathrm{M}}=\mathcal{G}\left(\boldsymbol{g}_l;\theta_{\mathrm{M}}\right),\tag{8}$$

where $\boldsymbol{g}_l=[x_l,y_l]^T$ represents the two-dimensional coordinates of the UE at the lth location, and θ_{M} is the set of the parameters of the radio map.

The normalized location of UE with the range of $(0,1)$ is fed to the input layer, and ReLU is applied here to activate neurons. Then, multiple hidden layers are utilized to expand the dimension of the input data and learn the relationship between the position and the beamforming vector. Next, the output layer applies tanh activation function to scale the output to $(-1,1)$ range. After that, the result of the output layer can be reshaped to obtain the real and imaginary parts of the beamforming vector, which can be combined to form the complex beamforming vector. Finally, normalization is performed to meet the power constraint.

During the training process, the same loss function for the reduced pilots is used, and the parameters can be obtained by minimizing the loss function defined in (6), i.e.,

$$\theta_{\mathrm{M}}^{*}=\arg\min_{\theta_{\mathrm{M}}}\frac{1}{L}\sum_{l=1}^{L}\mathcal{L}\Big(\mathcal{G}\left(\boldsymbol{g}_l;\theta_{\mathrm{M}}\right),\mathbf{H}_l\Big).\tag{9}$$

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B. Integration Scheme

The integrated beamforming vector $\mathbf{V}_l^{\mathrm{I}}$ of the UE at the lth location is acquired by fusing the beamforming vectors $\mathbf{V}_l^{\mathrm{M}}$ and $\mathbf{V}_l^{\mathrm{P}}$ generated by radio map and reduced pilots, respectively, i.e.,

$$\mathbf{V}_l^{\mathrm{I}} = \mathcal{H}\left(\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{V}_l^{\mathrm{P}}\right); \theta_{\mathrm{I}}\right), \tag{10}$$

where θ_{I} is the set of the parameters of the neural network for integration.

The framework of the integration scheme is shown in Fig. 2. In the figure, the beamforming vector V^{M} generated by the radio map and V^{P} obtained from reduced pilots are decomposed into real and imaginary parts. These decomposed parts are then concatenated to form a tensor of shape $(K, M, 4)$, which contains high-dimensional representations of location and pilot information. CNN is employed with a convolutional kernel size of 3×3, and both stride and zero-padding are set to 1 to ensure the size remains unchanged. Similar to the structure in the reduced pilots-based beamforming as shown in Fig. 1, two convolutional layers are used to extract more features by increasing the volume from $(K, M, 4)$ to $(K, M, 32)$. ReLU is adopted as the activation function here. Subsequently, these features are aggregated to a $(K, M, 2)$ tensor, and tanh is used to make the output range within $(-1, 1)$. After that, two channels of the tensor are combined for the complex value, and normalization is performed to obtain the integrated beamforming vector V^{I}.

In the training part, the parameters for the integration scheme can be yielded by minimizing the loss function as

$$\theta_{\mathrm{I}}^* = \arg\min_{\theta_{\mathrm{I}}} \frac{1}{L}\sum_{l=1}^{L} \mathcal{L}\Big(\mathcal{H}\left(\left(\mathbf{V}_l^{\mathrm{M}}, \mathbf{V}_l^{\mathrm{P}}\right); \theta_{\mathrm{I}}\right), \mathbf{H}_l\Big), \tag{11}$$

V. Numerical Simulations and Results

In this section, we present the chosen dataset and provide detailed parameters for numerical simulations. Then, simulation results are illustrated, and discussions are provided.

A. Dataset and Parameters

In the numerical simulations, the DeepMIMO dataset [14] generated by Remcom Wireless InSite [15] is adopted. The frequency f_c is 3.5GHz, and the subcarrier spacing is set to 15kHz. The number of subcarriers is 624, which consists of 52 resource blocks, and we focus on the first resource block, i.e., $K = 12$. We add complex Gaussian noise to the generated CSI, ensuring an average SNR of 30dB across all sampled locations. The uniform linear array (ULA) is applied by the BS for simplification, and the number of transmit antennas M is 32. We consider four pilot configurations with $\eta = 100\%$, 75%, 50%, and 25%, which indicates the ratios of antennas transmitting pilots to all 32 antennas. The corresponding proportion of the pilot overhead ρ for these four configurations are $9.52\%, 7.14\%, 4.76\%$, and 2.38%, respectively.

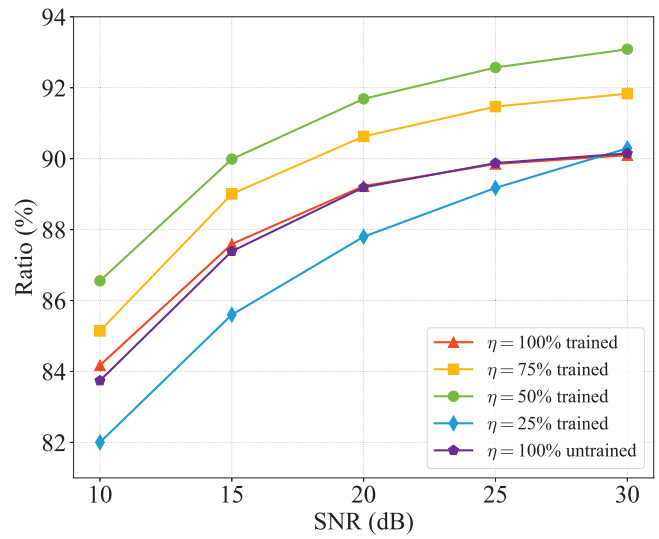

Fig. 3: The performances of the reduced pilots-based beamforming and full pilots versus SNR.

B. Numerical Results

After taking pilot overhead into consideration, we display the performances of the trained reduced pilots-based beamforming for these four configurations of pilots versus SNR in Fig. 3. The untrained full pilot configuration is applied as a baseline for comparison. The X-axis represents the average SNR for all sampled locations, and the Y-axis represents the ratio of the SE achieved by the method relative to the optimal SE with perfect CSI knowledge. In the figure, the five curves all increase with SNR, where $\eta = 50\%$ achieves the best performance among them, followed by $\eta = 75\%$. This suggests that the concept of reduced pilots is meaningful. Although the accuracy of channel estimation decreases with reduced pilots, the saved time-frequency resources can be used for more valuable data, which makes the actual SE higher than that with full pilots. In summary, $\eta = 50\%$ achieves the best tradeoff between beamforming performance and pilot overhead and will be adopted for integration with the radio map subsequently .

To validate the trained radio map, we apply GBM [16] and CKM method [11] as benchmarks. The performances of these two benchmarks and radio map method under LoS and NLoS conditions versus SNR are shown in Fig. 4. Although the performance at different locations is presented separately according to LoS and NLoS conditions, it should be clarified that during the training of radio map, we do not train two separate models for these two conditions. Instead, a unified model is trained on the mixed data. As shown in Fig. 4, GBM method performs best in LoS condition while worst in NLoS condition. This is because in LoS condition, the angle calculated based on the positions of BS and UE corresponds to the actual signal transmission angle. However, the situation is quite different in NLoS conditions. The beam is focused on obstacles, which greatly reduces the communication performance. The radio

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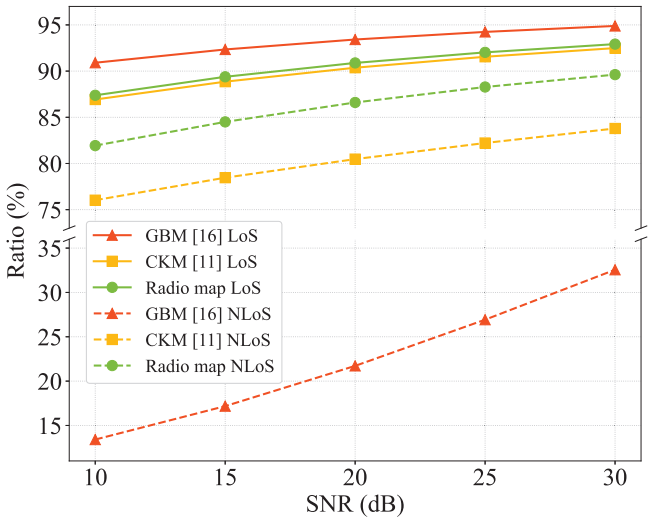

Fig. 4: The performances of GBM, CKM, and radio map methods under LoS and NLoS conditions versus SNR.

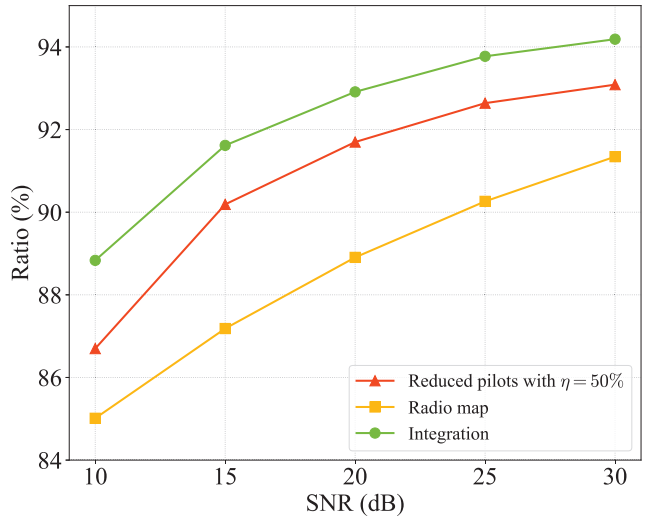

Fig. 5: The performances of integration scheme versus SNR.

map method slightly outperforms the CKM method in LoS condition but is significantly better in NLoS condition. This clarifies that deep learning-based methods have a distinct advantage over traditional methods like CKM, particularly in complex scenarios such as NLoS situations. Taking both LoS and NLoS conditions into consideration, the radio map-based beamforming method is superior to both GBM and CKM methods, exhibiting better applicability in different scenarios.

In Fig. 5, we show the performances of integration scheme versus SNR. To measure the improvement brought by integration scheme, reduced pilots-based beamforming and radio map-based beamforming are applied as baselines for comparison. As shown in Fig. 5, the red curve represents the reduced pilots-based method, which corresponds to the green curve in Fig. 3 and is the best among the four pilot configurations. Compared with radio map, reduced pilots-based method outperforms the radio map approach. In summary, the proposed integration scheme successfully integrates these two methods, combining the robustness of the radio map with the high performance of pilots at high SNRs.

VI. Conclusion

In this paper, we first propose a reduced pilots-based beamforming method and investigate the trade-off between pilot overhead and beamforming performance. Then, an integration scheme of reduced pilots and radio map is proposed to combine their advantages and compensate for shortcomings. The proposed scheme in this paper successfully integrates radio map and reduced pilots, achieving a balance between pilot overhead and beamforming performance and serving as a promising solution for emergency communications.

References

[1] Y. Ran, "Considerations and suggestions on improvement of communication network disaster countermeasures after the Wenchuan earthquake," *IEEE Commun. Mag.*, vol. 49, no. 1, pp. 44–47, Jan. 2011.

[2] E. K. Markakis *et al.*, "Efficient next generation emergency communications over multi-access edge computing," *IEEE Commun. Mag.*, vol. 55, no. 11, pp. 92–97, Nov. 2017.

[3] Y. Sun and K. R. Chowdhury, "Enabling emergency communication through a cognitive radio vehicular network," *IEEE Commun. Mag.*, vol. 52, no. 10, pp. 68–75, Oct. 2014.

[4] J. Wang, Y. Wu, N. Yen, S. Guo, and Z. Cheng, "Big data analytics for emergency communication networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 3, pp. 1758–1778, 3rd Quart. 2016.

[5] M. Berioli, A. Molinaro, S. Morosi, and S. Scalise, "Aerospace communications for emergency applications," *Proc. IEEE*, vol. 99, no. 11, pp. 1922–1938, Sep. 2011.

[6] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive MIMO for next generation wireless systems," *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 186–195, Feb. 2014.

[7] S. Shang, D. Xu, P. Ren, K. Yu, and M. Guizani, "Energy-efficient beam training for RIS assisted UAV communications in emergency rescue scenarios," in *Proc. IEEE Veh. Technol. Conf. (VTC-Spring)*, Mar. 2023, pp. 1–5.

[8] C. Wang, D. Deng, L. Xu, and W. Wang, "Resource scheduling based on deep reinforcement learning in UAV assisted emergency communication networks," *IEEE Trans. Commun.*, vol. 70, no. 6, pp. 3834–3848, Jun. 2022.

[9] R. Ferrus, O. Sallent, G. Baldini, and L. Goratti, "Public safety communications: Enhancement through cognitive radio and spectrum sharing principles," *IEEE Veh. Technol. Mag.*, vol. 7, no. 2, pp. 54–61, Jun. 2012.

[10] S. Bi, J. Lyu, Z. Ding, and R. Zhang, "Engineering radio maps for wireless resource management," *IEEE Wireless Commun.*, vol. 26, no. 2, pp. 133–141, Apr. 2019.

[11] Y. Zeng and X. Xu, "Toward environment-aware 6G communications via channel knowledge map," *IEEE Wireless Commun.*, vol. 28, no. 3, pp. 84–91, Jun. 2021.

[12] W. Wang, B. Yang, and W. Zhang, "Deep learning-based radio map for MIMO-OFDM downlink precoding," *J. Commun. Inf. Netw.*, vol. 8, no. 3, pp. 203–211, Sep. 2023.

[13] B. Yang, W. Wang, and W. Zhang, "Cell-free massive MIMO beamforming based on radio map," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Jun. 2024, pp. 1–6.

[14] A. Alkhateeb, "DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications," in *Proc. IEEE Inf. Theory Appl. Workshop (ITA Wkshps)*, Feb. 2019, pp. 1–8.

[15] Remcom, "Wireless InSite," http://www.remcom.com/wireless-insite.

[16] P. Kela *et al.*, "Location based beamforming in 5G ultra-dense networks," in *Proc. IEEE Veh. Technol. Conf. (VTC-Fall)*, Sep. 2016, pp. 1–7.

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